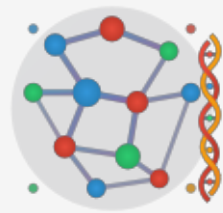


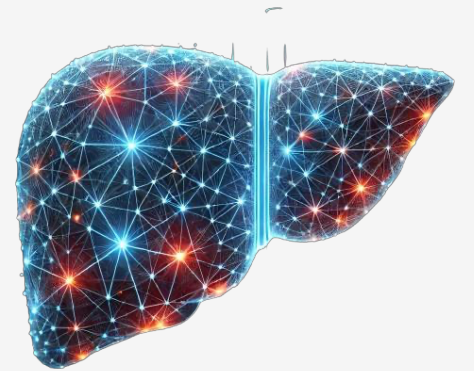
Transcriptomics Data Analysis



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What is in this lecture?

By the end, students can explain

- What RNA-seq measures
- Why design and QC matter
- What normalization does and does not do
- How PCA, differential expression and enrichment connect
- What can go wrong before interpretation

Key message: RNA-seq analysis is less about pushing buttons and more about deciding what comparison is scientifically valid.

Class Rhythm

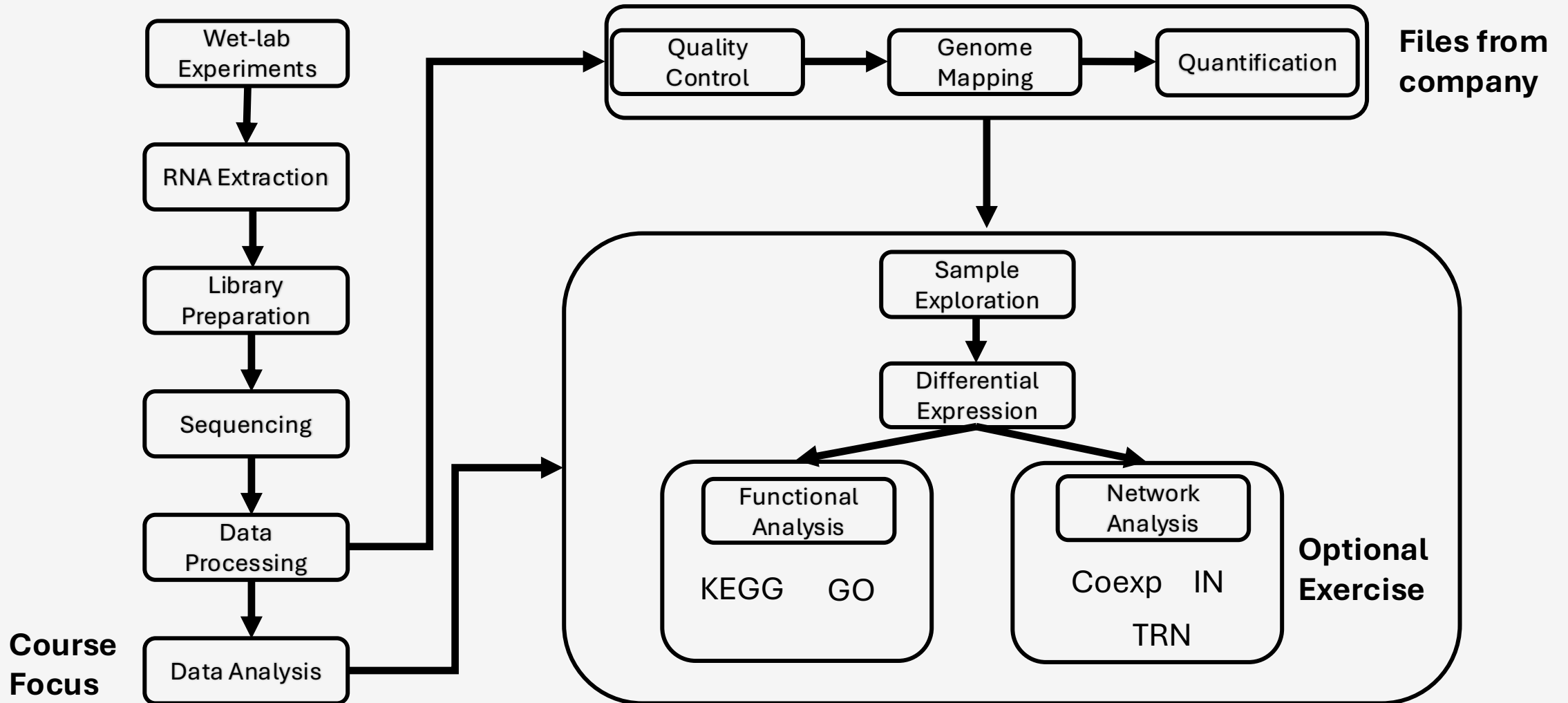
Short conceptual Blocks

- Quick activity
- Interpretation question
- Link to notebook

Note

- I will not discuss the experimental part extensively
 - i.e. batch effect, randomization, RIN score, confoundings etc
- You guys know about this much better than I do :D
- I will also not discuss about the conversion from FASTQ to Count files, most companies provide this now
 - Also for the sake of time
 - Hint: Kallisto, Salmon → my favorite tools for quantification
 - Drop me an email if you need to do this and I can guide you.

Transcriptomics Analysis Workflow

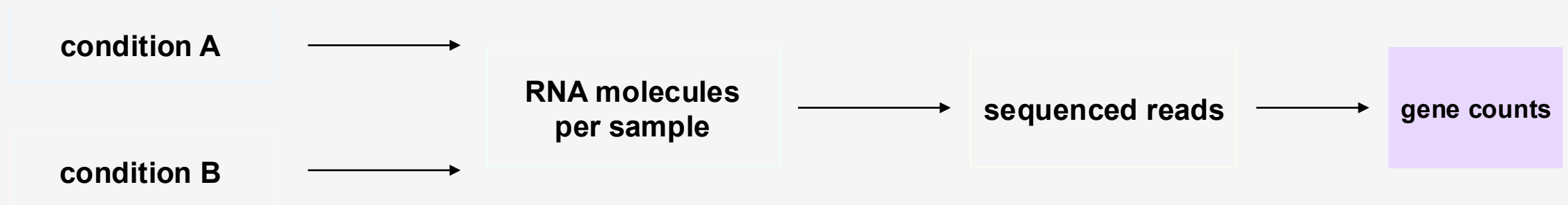


Learn the lingo!

- Knowing the lingo is important
- Counts, TPM, Metadata, Design, PCA, Contrasts, FDR, Enrichment etc etc
- Knowing them will make your life easier (i.e. easier to google something :D)

Data-Driven or Hypothesis-Driven?

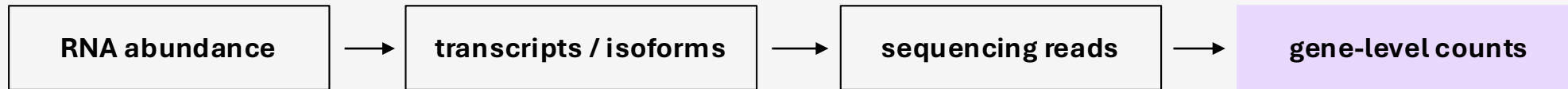
- It does not matter!
- Both requires proper biological question(s)
 - Before using any tools, ask yourself what do you want to interpret from the data
 - Exploration? Testing a hypothesis? Understanding drug effect?



Where can bias enter this pipeline before we even open the notebook?

What does RNA-seq measure?

- A count table is a compressed summary of a much richer molecular process



Gene-level analysis

Most beginner RNA-seq analyses collapse transcripts into one number per gene. This is enough for many biological questions.

Transcript-level analysis

Alternative splicing and isoform usage require more specialized design, tools, and interpretation.

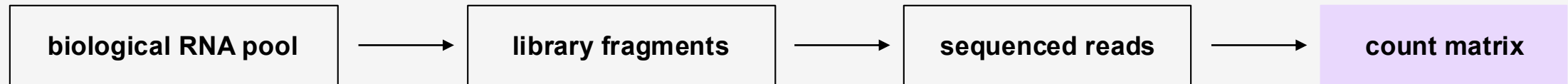
Important limitation

Counts are not absolute molecule numbers. They are observed reads after sampling and processing.

For today: we focus on gene-level differential expression from a count matrix.

RNA-seq is a sampling experiment

- Reads are a sample from a complex pool of RNA molecules



Sampling depth

More reads usually means better chance to detect low-abundance transcripts, but it does not replace biological replicates.

Composition effect

If a few genes dominate one sample, the relative space for all other genes changes.

Annotation choice

Counts depend on the genome build and gene model used to assign reads.

**RNA-seq does not count molecules directly.
It counts reads that survived many biological and technical steps.**

What biological changes can RNA-seq detect?

More cells of one type

A tissue sample may show higher immune genes because there are more immune cells, not because every cell changed transcription.

More RNA per cell

A cell state may increase transcription of specific genes within the same cell population.

Different isoform usage

Gene-level counts may hide transcript switches if total gene abundance stays similar.

Discuss!

A fibrosis sample has higher collagen genes. Which explanation is most plausible?

Expression is relative and contextual

- A gene can be statistically changed, biologically tiny, or biologically meaningful but uncertain

Effect size

How large is the change? In RNA-seq this is usually reported as log2 fold change.

Evidence

How consistent is the change across biological replicates?

Biological context

Is the gene expressed in the relevant tissue or pathway? Does the direction make sense?

scenario	log2FC	expression	replicates	interpretation
large shift, noisy	2.5	low	few	interesting but uncertain
tiny shift, precise	0.15	high	many	statistical, maybe not biological
moderate shift, consistent	1.0	medium	balanced	good candidate

**Advice: Do not use P-Value as your only criterion.
Use your biological knowledge to put context to the data**

Activity: Count-table detective

- Two minutes in pairs: what can and cannot be concluded from this table?

gene	A1	A2	B1	B2
Gene X	500	520	950	980
Gene Y	20	22	45	41
Gene Z	10,000	9,800	11,200	10,900
Total reads	18M	19M	41M	40M

Which genes look higher in condition B?

Which answer changes after considering total reads?

What extra information do we need before doing statistics?

Raw counts mix **biology + library size + technical effects.**

Why raw counts are not directly comparable

- More reads can create bigger counts without more RNA

gene	A1	A2	B1	B2
Gene X	500	520	950	980
Gene Y	20	22	45	41
Gene Z	10,000	9,800	11,200	10,900
Total reads	18M	19M	41M	40M

At first glance: Gene X & Y looks higher in B.

But B was sequenced roughly twice as deeply.

Normalization asks:

(TPM for this course, other: RPKM, FPKM, CPM)

What part of the count difference is due to sequencing depth or composition, rather than biology?

Common outputs

normalized counts for plots

TPM for transcript abundance summaries

model-based size factors for differential expression

**Do not mix goals: TPM is useful for expression summaries/Visualization
Counts are used for differential testing (Internal tool normalization).**

Short Break!

Next: More practical interpretation
PCA, Volcano Plots, Enrichment

Sample exploration: PCA is a question generator

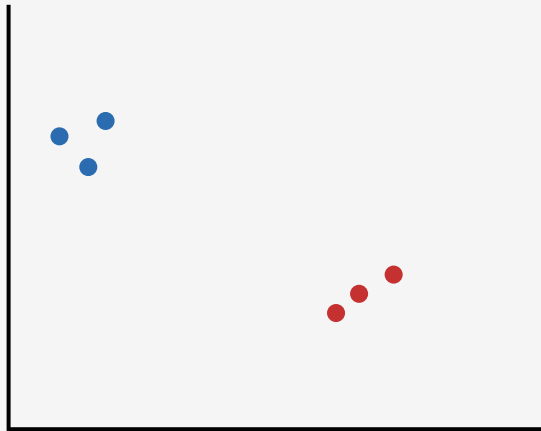
- PCA shows major axes of variation; it does not explain them automatically

Do samples cluster by biology?

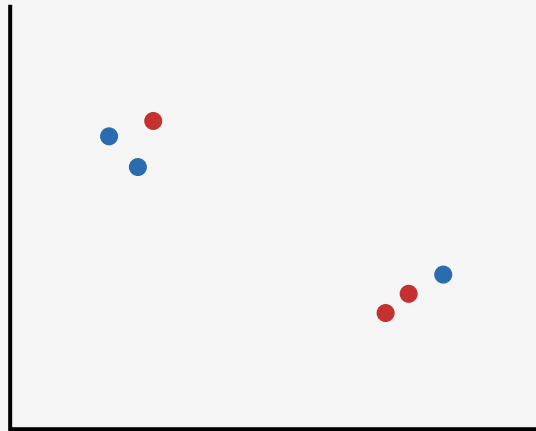
Do they cluster by batch, lane, sex, or RIN?

Are there outliers or possible sample swaps?

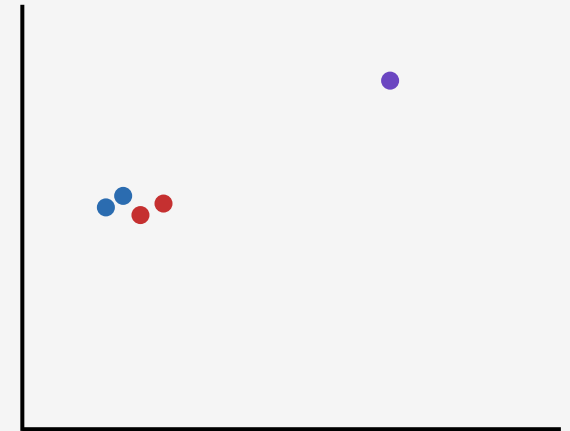
Clear condition structure



Pattern not matching condition



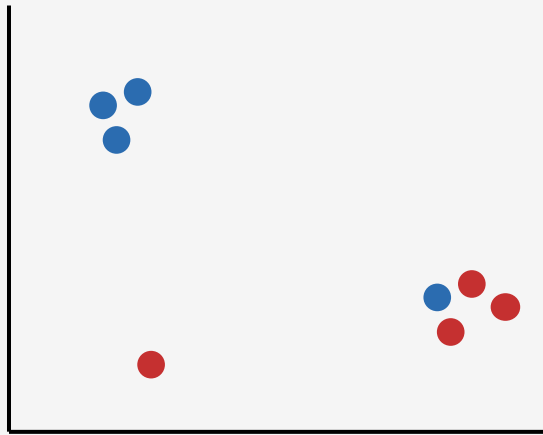
Possible outlier



PCA should send you back to the metadata.

Activity: Mystery PCA

- What is the most likely explanation?



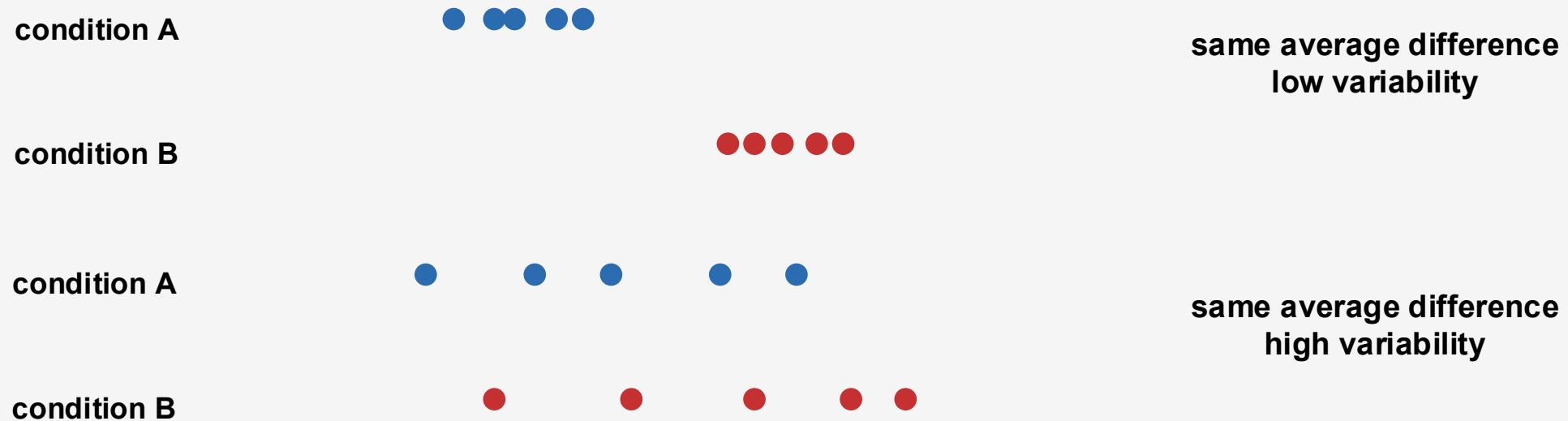
- A: condition is the main driver
- B: batch effect dominates
- C: one sample is swapped
- D: all samples are fine; ignore it

Follow-up: Which metadata columns would you color by next?

there is no single answer from PCA alone. PCA is a triage plot.

Replicates estimate biological variability

- Differential expression is signal compared with variation



Technical replicates estimate measurement noise.
Biological replicates estimate person/animal/culture variability.
More replicates improve uncertainty estimates.

why RNA-seq is not just many t-tests

- Counts are discrete, skewed, and their variance depends on their mean

Low counts

0, 1, 2, 0, 3

A small absolute change can look large as a fold change.

High counts

900, 980, 1050, 970

Large absolute differences may be small relative changes.

Overdispersion

Biological variation is often larger than simple counting noise would predict.

DESeq2, edgeR, and similar methods model counts and estimate gene-wise variability using information across genes.

The model asks: is the difference between groups larger than expected variation?

The answer depends on count level, replicate variability, and design formula.

This is why the same log₂FC can be convincing for one gene and weak for another.

many genes mean many chances to be “fooled”

- False discovery rate controls the expected fraction of false positives among discoveries

tests	$p < 0.05$ false positives expected by chance
1 gene	0.05
100 genes	about 5
20,000 genes	about 1,000

p-value

Evidence against the null hypothesis for one test.

adjusted p-value / FDR

Evidence after accounting for the number of genes tested.

Practical habit: discuss genes using both log2FC and FDR, then bring in expression level and biology.

Reading differential-expression output

- Fold change, p-value, and adjusted p-value answer different questions

gene	log2FC	p-value	FDR	interpretation
A	2.1	0.040	0.31	large but uncertain
B	0.18	1e-8	0.002	tiny but reliable
C	1.1	1e-6	0.004	good follow-up candidate

log2 fold change

Magnitude and direction of change.

+1 = about 2× higher.

−1 = about 2× lower.

p-value

How surprising the data are under the null hypothesis for one gene.

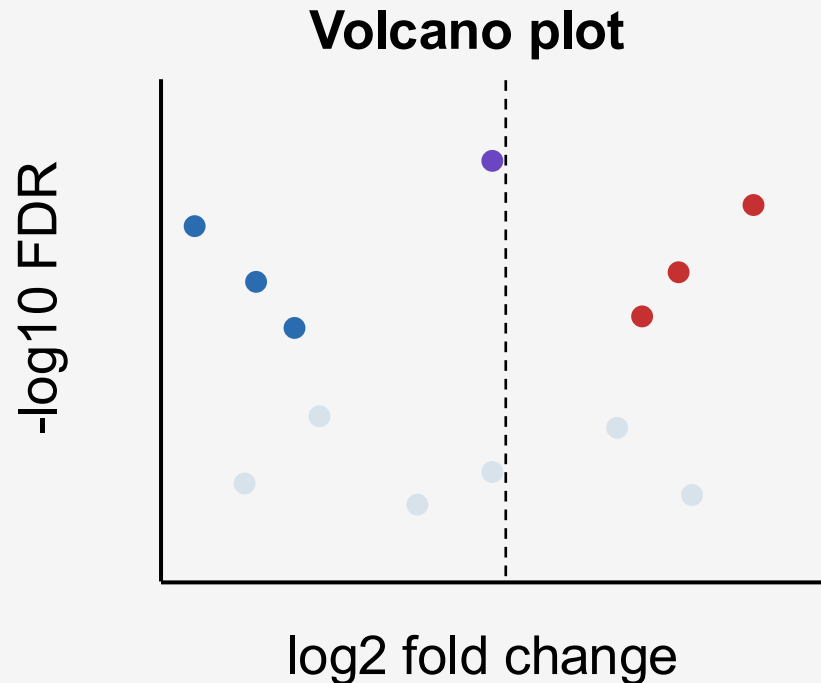
FDR / adjusted p-value

Controls expected false discoveries across many tested genes.

A gene is interesting because of biology + effect size + uncertainty + context, not one number alone.

Volcano plots: good overview, bad for shopping list

- The plot combines effect size and statistical evidence



Left/right

Direction of change. Left = lower in condition B; right = higher in condition B, depending on contrast.

Up/down

Statistical evidence. Higher points have smaller adjusted p-values.

Danger

The most extreme point is not always the best biological story. Check expression level, annotation, and consistency.

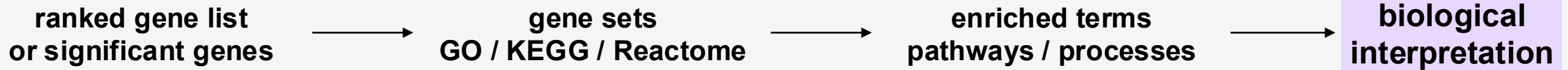
Volcano plot = map for discussion, not automatic truth.

Activity: which gene would you follow up?

gene	known role	mean count	log2FC	FDR
Gene A	unknown	12	3.2	0.08
Gene B	pathway of interest	600	0.55	0.001
Gene C	marker gene	80	1.4	0.02
Gene D	housekeeping	20,000	0.15	1e-6

From genes to functions

- Differential expression gives gene-level evidence; enrichment asks about biological programs



ORA

Over-representation analysis asks whether a selected gene list contains more pathway members than expected.

GSEA

Gene-set enrichment analysis uses a ranked list and can detect coordinated small changes.

Functional analysis should summarize biology, not decorate a gene list.

Common enrichment traps

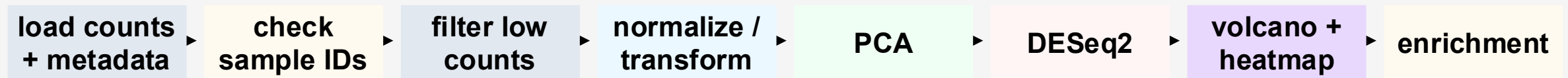
Most errors are interpretation errors, not software errors

- Wrong background: using all genes in the genome instead of all genes tested.
- Direction ignored: up-regulated and down-regulated genes mixed together.
- Redundant terms: many GO terms say almost the same thing.
- Annotation bias: well-studied pathways look more convincing.
- Overclaiming causality: enrichment suggests association, not mechanism by itself.

**Enrichment is a hypothesis generator.
Connect them back to the biology.**

What the hands-on will do

- Turning concept into reproducible analysis workflow



Optional: Multivariable DESeq2 (Confounding)

Every step has a question

Do samples match? Are low-information genes removed? Are samples grouped as expected? Which contrast is tested?

Every plot needs interpretation

PCA, volcano plots, heatmaps and enrichment results are not endpoints. They are evidence for a biological story.

The notebook is the “how”; this lecture is the “why”.

Your own analytical checklist

- Before trusting your analysis, ask these questions
 - Is the biological contrast clear?
 - Are replicates and batches interpretable?
 - Do metadata and sample IDs match?
 - Do QC and PCA support the design?
 - Was normalization appropriate for the task?
 - Do fold change, FDR and expression level agree?
 - Does functional analysis match known biology without overclaiming?

Good analysis = valid design + careful QC + honest interpretation



A word cloud featuring the phrase "Thank You" in numerous languages. The central and largest text is "thank you" in a bold, red, sans-serif font. Surrounding this are many other languages, including Spanish ("gracias", "gracias", "gracias"), French ("merci", "merci", "merci"), German ("danke", "danke", "danke"), Italian ("grazie", "grazie", "grazie"), Japanese ("ありがとう", "ありがとう", "ありがとう"), Chinese ("谢谢", "谢谢", "谢谢"), Hindi ("धन्यवाद", "धन्यवाद", "धन्यवाद"), and many others. The words are arranged in a circular pattern, with some words appearing multiple times in different colors and sizes. The background is white.

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BioJupies

- Web-based automated RNA-seq Data Analysis Notebook
- Easier, but less flexible

<https://maayanlab.cloud/biojupies/>

The screenshot shows the BioJupies website interface. At the top, there is a search bar for GEO datasets, links for Chrome Extension, Help, and a Sign In button. The main content area features a three-step process for generating RNA-seq analysis notebooks:

- Step 1. Upload or Fetch RNA-seq Data**
 - Upload your raw or processed RNA-seq data
 - Fetch >8,000 public RNA-seq datasets published in the Gene Expression Omnibus
- Step 2. Select Data Analysis Tools**
 - Select from multiple state-of-the-art RNA-seq data analysis tools
 - Contribute your computational tool as a plugin
- Step 3. Generate Your Notebook**
 - Access and share your results through a permanent URL
 - Download, rerun and customize your notebook using Docker

To the right of the steps, the text reads: "BioJupies Automatically Generates RNA-seq Data Analysis Notebooks". Below this, it states: "With BioJupies you can produce in seconds a customized, reusable, and interactive report from your own raw or processed RNA-seq data through a simple user interface". A "Get Started" button is located at the bottom right of this section.

To acknowledge BioJupies in your publications, please use the following reference: Torre, D., Lachmann, A., and Ma'ayan, A. (2018). BioJupies: Automated Generation of Interactive Notebooks for RNA-Seq Data Analysis in the Cloud. Cell Systems.